SCALABILITY & FUTURE PLAN

What

The Automated Network Request Management solution is designed with a scalable foundation that can be extended as the organization's network and IT service requirements grow.

The current implementation establishes the core capabilities for request submission, data capture, approval, automation, notification, and tracking. Future enhancements can extend these capabilities to provide deeper infrastructure automation, advanced reporting, and broader IT service management.

Why

As the number and complexity of network requests increase, additional automation and monitoring capabilities can help the organization:

- Reduce manual infrastructure operations.
- Improve request fulfilment speed.
- Handle larger request volumes.
- Provide better operational visibility.
- Integrate ServiceNow with infrastructure automation platforms.
- Standardize service delivery across multiple IT services.
- Reduce repetitive administrative activities.
- Support future digital transformation initiatives.

How

Automated Network Provisioning

A future enhancement is to connect ServiceNow with network automation platforms such as Cisco DNA Centre or Ansible.

After the required approvals are completed, ServiceNow could send the approved request information to the appropriate automation platform.Potential use cases include:

- Network configuration provisioning.
- VLAN creation.
- Device configuration.
- Connectivity provisioning.
- Configuration updates.
- Automated network change execution.

This would reduce the need for network engineers to manually perform repetitive configuration activities.

Future Enhancement: This integration is not part of the current implementation and can be introduced as a later development phase.

ServiceNow Integration Hub

ServiceNow Integration Hub can be introduced to extend the solution beyond ServiceNow's internal automation capabilities.

Integration Hub could be used to connect ServiceNow with external enterprise systems and automation services.Potential future integrations include:

- Network management platforms.
- Infrastructure automation tools.
- Identity and access management systems.
- Asset management systems.
- Monitoring platforms.

This would allow approved Network Requests to participate in broader enterprise automation processes.

Reporting and Dashboards

Future versions can introduce dedicated dashboards to provide management and operational visibility into Network Requests. Potential dashboard metrics include:

Metric	Purpose
Total Requests	Measures overall request volume
Pending Requests	Identifies requests awaiting processing or approval
Approved Requests	Tracks successfully approved requests
Rejected Requests	Identifies rejected requests
Fulfilled Requests	Measures completed requests
Average Fulfilment Time	Measures operational efficiency
Requests by Type	Identifies frequently requested services
Requests by Department	Shows demand across departments
Approval Turnaround Time	Measures approval efficiency

These dashboards can help administrators and management identify bottlenecks and make data-driven decisions.

Expansion to Other IT Services

The same architecture can be extended beyond network requests.

Potential future catalog services include:

- Device Provisioning
- Access Requests
- Software Installation Requests
- Hardware Requests
- Account Provisioning
- Infrastructure Requests
- IT Equipment Requests

The existing approach of structured catalog items, variables, approvals, automation, notifications, and tracking can be reused and adapted for these services.

Scalability Strategy

The future scalability of the solution can be considered across four major areas:

Area	Current Capability	Future Enhancement
Request Management	Network Request catalog item	Multiple IT service catalog items
Automation	ServiceNow Flow Designer	Infrastructure and external-system automation
Integration	ServiceNow-native processing	Integration Hub and network-tool integrations
Analytics	Request tracking	Advanced operational dashboards and KPIs

Future Development Roadmap

The recommended future enhancement sequence is:

Phase 1 – Current Solution
Network Request submission, validation, approval, automation, notification, and tracking.

Phase 2 – Advanced Reporting
Introduce dashboards and operational KPIs.

Phase 3 – Enterprise Integration
Introduce Integration Hub and connections to external infrastructure systems.

Phase 4 – Auto-Provisioning
Enable approved requests to trigger automated network configuration.

Phase 5 – Service Expansion
Extend the solution to device provisioning, access requests, and other IT services.

Outcome

The solution provides a foundation that can evolve from a network request management application into a broader IT service automation platform.

By progressively introducing integrations, infrastructure automation, analytics, and additional service catalog offerings, the organization can increase automation without replacing the existing request-management foundation.

Replicability & Maintainability Note

The current solution should remain the baseline implementation, while future capabilities can be introduced incrementally. This approach reduces implementation risk and allows new integrations, services, and reporting capabilities to be added without redesigning the complete request-management process.

Important Documentation Note

The following items should be presented in the capstone report as future enhancements, not as currently implemented features:

- Cisco DNA Centre integration
- Ansible integration
- Integration Hub orchestration
- Automated network provisioning
- Advanced reporting dashboards
- Expansion to additional IT services